

HW 11

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Townsend: 11.15, 12.2, 12.4, 12.8

Townsend Problem 11.15 (5 pts)

Determine the effect of an external magnetic field on the energy levels of the $n = 2$ states of hydrogen when the applied magnetic field B has a magnitude much greater than 10^4 gauss, in which case the spin-orbit interaction may be neglected as a first approximation. This is the **Pashen-Bach effect**.

Draw an energy-splitting diagram like Figure 11.10.

Hint: Unlike the section problem and most of the section on the Zeeman Effect, you can start with the (11.105) form rather than the (11.106) form of the perturbing Hamiltonian.

We have four possible states for $n = 2$

$$|2, 0, 0\rangle, |2, 1, -1\rangle, |2, 1, 0\rangle, |2, 1, 1\rangle$$

each of which can have $S = \pm \frac{1}{2}$

We can find the energy of the perturbing Hamiltonian on each of those states.

$$\hat{H}_B = \frac{eB}{2m_e c} (\hat{L}_z + 2\hat{S}_z) \quad (11.105)$$

We can build the matrix to see how this applies:

$$\langle \psi | \hat{H}_B | \psi \rangle = \frac{eB}{2m_e c} \langle \psi | \hat{L}_z + 2\hat{S}_z | \psi \rangle$$

I'll call $\frac{eB}{2m_e c} = C$

We can write this out for every state

$$C(m_l + 2m_s)$$

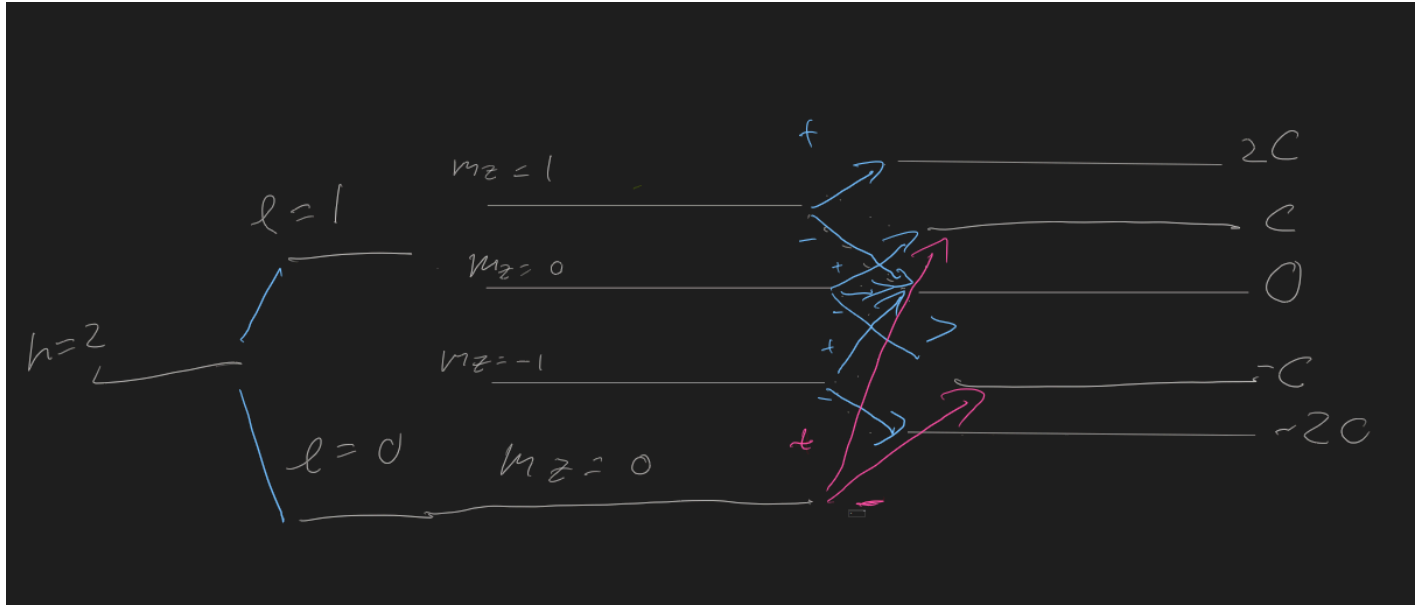
For all spin $\frac{1}{2}$, that will be $m_l + 1$, and all $-\frac{1}{2}$ it will be $m_l - 1$.

We get

$$\begin{aligned} |2, 0, 0, \frac{1}{2}\rangle &= C \\ |2, 0, 0, -\frac{1}{2}\rangle &= -C \\ |2, 1, -1, \frac{1}{2}\rangle &= 0 \\ |2, 1, -1, -\frac{1}{2}\rangle &= -2C \\ |2, 1, 0, \frac{1}{2}\rangle &= C \\ |2, 1, 0, -\frac{1}{2}\rangle &= -C \\ |2, 1, 1, \frac{1}{2}\rangle &= 2C \\ |2, 1, 1, -\frac{1}{2}\rangle &= 0 \end{aligned}$$

These are all the expectation values of energy, the diagonal elements of a matrix where the rows and columns are these as bras and kets, with the hamiltonian sandwiched in between. These are how much each of the eigenstate's energies are changed, and linear combinations of those eigenkets will have the same changes applied.

All of these have a root energy E_0 , from which they are $\pm 0, 1$, or $2c$



TODO find the energies with these

We know that

$$\Delta E = h\Delta\nu$$

We can plug this in to find the energy released by electrons going between the adjacent energy levels

Townsend Problem 12.2 (5 pts)

Two identical, noninteracting spin- $\frac{1}{2}$ particles of mass m are in the one-dimensional harmonic oscillator for which the Hamiltonian is

$$\hat{H} = \frac{\hat{p}_1^2}{2m} + \frac{1}{2}m\omega^2\hat{x}_1^2 + \frac{\hat{p}_2^2}{2m} + \frac{1}{2}m\omega^2\hat{x}_2^2$$

- (a) Determine the ground-state and first-excited-state kets and corresponding energies when the two particles are in a total-spin-0 state. What are the lowest energy states and corresponding kets for the particles if they are in a total-spin-1 state?
- (b) Suppose the two particles interact with a potential energy of interaction

$$V(|x_1 - x_2|) = \begin{cases} -V_0 & |x_1 - x_2| < a \\ 0 & \text{elsewhere} \end{cases}$$

Argue what the effect will be on the energies that you determined in (a), that is, whether the energy of each state moves up, moves down, or remains unchanged. *Suggestion:* Examine which spatial wave functions for the total-spin-0 and total-spin-1 states tend to have the particles closer together. Consider, for example, the special case of $x_1 = x_2$.

A

To have a total spin 0 state, the particles must have opposite spin.

The two particles are not interacting and spin isn't in the Hamiltonian, so they'll occupy the regular ground states.

The ground states of the SHO is a Gaussian, and both of them have the same Gaussian because they have the same Hamiltonian. Thus, the spatial ground state for each particle $x = 1$ and $x = 2$.

$$\langle r|0\rangle_x = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{3}{4}} e^{-m\omega\frac{x^2}{2\hbar}}$$

The first excited state for each is the same as the $|1\rangle$ of the SHO:

$$\langle r|1\rangle_x = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{2}} \left[\frac{4}{\pi} \left(\frac{m\omega}{\hbar}\right)^3\right]^{\frac{1}{4}} r \sin\theta \cos\phi e^{-\mu\omega\frac{r^2}{2\hbar}}$$

$$E_{nx} = \left(n + \frac{3}{2}\right)\hbar\omega$$

Both can be in their ground states (lowest energy) at the same time, since spin doesn't matter for any interaction (they have opposite spin, $\pm\frac{1}{2}$). Under exchange (because they are identical particles), we want them to not change, so the state must NOT be symmetric (fermions can't stack).

1. Spatial part:

$$\Phi_S(1,2) = \frac{1}{\sqrt{2}} [\psi_{1s}(\vec{r}_1)\psi_{2s}(\vec{r}_2) + \psi_{2s}(\vec{r}_1)\psi_{1s}(\vec{r}_2)]$$

$$\Phi_A(1,2) = \frac{1}{\sqrt{2}} [\psi_{1s}(\vec{r}_1)\psi_{2s}(\vec{r}_2) - \psi_{2s}(\vec{r}_1)\psi_{1s}(\vec{r}_2)]$$

2. Spin part:

$$\chi_S(1,2) = \chi_+(1)\chi_+(2)$$

$$\chi_S(1,2) = \chi_-(1)\chi_-(2)$$

$$\chi_S(1,2) = \frac{1}{\sqrt{2}} [\chi_+(1)\chi_-(2) + \chi_-(1)\chi_+(2)]$$

$$\chi_A(1,2) = \frac{1}{\sqrt{2}} [\chi_+(1)\chi_-(2) - \chi_-(1)\chi_+(2)]$$

Triplet States:

$$\Psi(1,2) = \frac{1}{\sqrt{2}} [\psi_{1s}(\vec{r}_1)\psi_{2s}(\vec{r}_2) - \psi_{2s}(\vec{r}_1)\psi_{1s}(\vec{r}_2)]\chi_+(1)\chi_+(2)$$

$$\Psi(1,2) = \frac{1}{\sqrt{2}} [\psi_{1s}(\vec{r}_1)\psi_{2s}(\vec{r}_2) - \psi_{2s}(\vec{r}_1)\psi_{1s}(\vec{r}_2)]\chi_-(1)\chi_-(2)$$

$$\Psi(1,2) = \frac{1}{2} [\psi_{1s}(\vec{r}_1)\psi_{2s}(\vec{r}_2) - \psi_{2s}(\vec{r}_1)\psi_{1s}(\vec{r}_2)][\chi_+(1)\chi_-(2) + \chi_-(1)\chi_+(2)]$$

The lowest energy state:

$$(|0\rangle_1 \otimes |0\rangle_2)(|+\rangle_1|-\rangle_2 + |-\rangle_1|+\rangle_2)$$

The next energy states are

symmetric spin

$$(|0\rangle_1 \otimes |1\rangle_2 + |1\rangle_1 \otimes |0\rangle_2)(|+\rangle_1|-\rangle_2 + |-\rangle_1|+\rangle_2)$$

antisymmetric spin

$$(|0\rangle_1 \otimes |1\rangle_2 + |1\rangle_1 \otimes |0\rangle_2)(|+\rangle_1|-\rangle_2 - |-\rangle_1|+\rangle_2)$$

$$(|0\rangle_1 \otimes |1\rangle_2 + |1\rangle_1 \otimes |0\rangle_2)(|+\rangle_1|-\rangle_2)$$

$$(|0\rangle_1 \otimes |1\rangle_2 + |1\rangle_1 \otimes |0\rangle_2)(|-\rangle_1|+\rangle_2)$$

B

If they are closer than some radius, then we have a square potential (it's a circle, a radial step function). We can act this new potential as a perturbation to the Hamiltonian - because it's a negative constant ($-v_0$), the energy of each particle ground state is just lowered by the constant when they are within a of each other.

The symmetric position means that under exchange of labeling, it's the same state - so the particles are in the same position. Yay FERMIONS!

Advice from Gallicchio:

make 2D plots (density of 2D wavefunction) 2D SHO

Ground state Gaussians (gauss in x * one in y , contour plot)

Townsend Problem 12.4 (5 pts)

Use the variational principle to estimate the ground-state energy for the one-dimensional anharmonic oscillator

$$\hat{H} = \frac{p_x^2}{2m} + bx^4$$

Compare your result with the exact result

$$E_0 = 1.060b^{1/3} \left(\frac{\hbar^2}{2m} \right)^{2/3}$$

We are doing a perturbation around the ground state of the SHO, $|0\rangle$, and vary the frequency. We don't know the eigenstates of the momentum because it's a free particle, so let's only consider the whole thing for the variation.
(thanks Annika)

the perturbing hamiltonian for the position term is

$$\langle 0|b\hat{x}^4|0\rangle$$

We can write this as the raising and lowering operators

We can't lower below the ground state, so we destroy any term ending in $\hat{a}$ or more lowering operators than raising in order, or any unequal numbers of raising and lowering.

$$\left(\frac{\hbar}{2m\omega}\right)^2 b \langle 0|(\hat{a} + \hat{a}^\dagger)^4|0\rangle$$

If we have equal raising and lowering, this is

$$\hat{a}^2\hat{a}^{2\dagger} + \hat{a}\hat{a}^\dagger\hat{a}\hat{a}^\dagger + \hat{a}\hat{a}^{2\dagger}\hat{a} + \hat{a}^{2\dagger}\hat{a}^2 + \hat{a}^\dagger\hat{a}\hat{a}^\dagger\hat{a} + \hat{a}^\dagger\hat{a}^2\hat{a}^\dagger$$

If we get rid of the other invalid ones, leaving the terms

$$\hat{a}^2\hat{a}^{2\dagger} + \hat{a}\hat{a}^\dagger\hat{a}\hat{a}^\dagger$$

We now get

$$\left(\frac{\hbar}{2m\omega}\right)^2 b * (2 + 1)$$

This gives the position part as

$$\boxed{3\left(\frac{\hbar}{2m\omega}\right)^2 b}$$

We can do the same with the momentum portion

$$\hat{p}_x = -i\sqrt{\frac{m\omega\hbar}{2}}(\hat{a} - \hat{a}^\dagger)$$

$$\hat{p}_x^2 = -\frac{m\omega\hbar}{2}(\hat{a}^2 + \hat{a}^{2\dagger} - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a})$$

We have the same rules applying, so we only have $-\hat{a}\hat{a}^\dagger$

$$\frac{\langle \hat{P}_x \rangle^2}{2m} = \frac{m\omega\hbar}{4m}$$

We combine these to get

$$E = 3\left(\frac{\hbar}{2m\omega}\right)^2 b + \frac{\omega\hbar}{4}$$

We can minimize this with respect to ω . Derive!

$$0 = -\frac{3\hbar^2 b}{4m^2} \frac{2}{\omega^3} + \frac{\hbar}{4}$$

$$\omega^3 = 6\hbar \frac{b}{m^2}$$

$$\omega = \sqrt[3]{6\hbar \frac{b}{m^2}}$$

We can use this to find E_0 :

$$E_0 = \frac{3b\hbar^2}{4m^2\omega^2} + \frac{\hbar\omega}{4}$$

$$= \frac{3b\hbar^2}{4m^2} \left(\frac{m^2}{6\hbar b} \right)^{2/3} + \frac{\hbar}{4} \left(\frac{6\hbar b}{m^2} \right)^{1/3}$$

This is going to be awful

$$\left(\frac{3}{4 \cdot 6^{2/3}} + \frac{6^{1/3}}{4} \right) b^{1/3} \frac{\hbar^{4/3}}{m^{2/3}}$$

This is

$$1.081 b^{1/3} \left(\frac{\hbar^2}{2m} \right)^{2/3}$$

The exact result is really close!

$$E_0 = 1.060 b^{1/3} \left(\frac{\hbar^2}{2m} \right)^{2/3}$$

Old work but still valid, just grosser

We don't have an ω term, and x^4 looks a lot like x^2 . We can take the guess that the wave function looks like a Gaussian, which is the ground state for x^2 , and vary sigma to approximate.

$$\langle \psi_\sigma | \hat{H} | \psi_\sigma \rangle > E_0$$

We can use the position space Gaussian wave packet,

$$\psi(x) = \frac{1}{\sqrt{\sqrt{\pi}\sigma}} e^{-\frac{x^2}{2\sigma^2}}$$

We can use the variational principle on σ .

The expectation value in this position space representation is

$$\frac{1}{\sqrt{\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}} \left(\frac{p_x^2}{2m} + bx^4 \right) e^{-\frac{x^2}{2\sigma^2}} dx$$

We can use the derivative version of momentum

$$\frac{1}{\sqrt{\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}} \left(\frac{-1}{2m} \hbar^2 \frac{d^2}{dx^2} + bx^4 \right) e^{-\frac{x^2}{2\sigma^2}} dx$$

$$\frac{1}{\sqrt{\pi}\sigma^3} \left(-\frac{\hbar^2}{2m} \right) \int_{-\infty}^{\infty} e^{-\frac{x^2}{2\sigma^2}} \left(\frac{x}{2\sigma^4} + \frac{e^{-\frac{x^2}{2\sigma^2}}}{\sigma^2} + bx^4 e^{-\frac{x^2}{2\sigma^2}} \right) dx$$

$$\frac{1}{\sqrt{\pi}\sigma^3} \left(-\frac{\hbar^2}{2m} \right) \int_{-\infty}^{\infty} \frac{x}{2\sigma^4} e^{-\frac{x^2}{2\sigma^2}} + \frac{1}{\sigma^2} e^{-\frac{x^2}{2\sigma^2}} + bx^4 e^{-\frac{x^2}{2\sigma^2}} dx$$

Lets break this up into three integrals

Integral 1

$$\int_{-\infty}^{\infty} \frac{x}{2\sigma^4} e^{-\frac{x^2}{2\sigma^2}} dx$$

We can u sub - let $u = x^2, du = 2x dx$

$$\frac{1}{4\sigma^4} \int_{-\infty}^{\infty} e^{\frac{-u}{2\sigma^2}} du = \frac{1}{8\sigma^6} e^{\frac{-u}{2\sigma^2}} \Big|_{-\infty}^{\infty}$$

infinity - zero?

integral 2

$$\int_{-\infty}^{\infty} \frac{1}{\sigma^2} e^{\frac{-x^2}{\sigma^2}} dx = \frac{1}{\sigma^4} e^{\frac{-x^2}{\sigma^2}} \Big|_{-\infty}^{\infty}$$

also infinity - zero?

integral three

$$\int_{-\infty}^{\infty} bx^4 e^{\frac{-x^2}{\sigma^2}} dx$$

We can use the derivative trick. Lets call $a = \sigma^2$

$$\begin{aligned} & b \int_{-\infty}^{\infty} x^4 e^{\frac{-x^2}{a}} dx \\ &= b \frac{d}{da} \int_{-\infty}^{\infty} -x^2 e^{\frac{-x^2}{a}} dx \\ &= b \frac{d^2}{da^2} \int_{-\infty}^{\infty} e^{\frac{-x^2}{a}} dx \\ &= b \frac{d^2}{da^2} e^{\frac{-x^2}{a}} \end{aligned}$$

This gives the energy as

$$\frac{1}{\sqrt{\pi}\sigma^3} \left(-\frac{\hbar^2}{2m} \right)$$

We want to find where this energy is minimized to get the ground state, so

Townsend Problem 12.8 (5 pts)

Use the trial wave function $R(r) = Ne^{-br}$ to estimate the ground-state energy of the three-dimensional isotropic harmonic oscillator, for which $V(r) = \frac{1}{2}\mu\omega^2 r^2$. Compare your estimate with the exact value (see Section 10.5).

We have the Hamiltonian

$$\hat{H} = \frac{\hat{P}^2}{2\mu} + \frac{1}{2}\mu\omega^2 \hat{r}^2$$

We guess that $R(r) = Ne^{-br}$

We can determine N , the normalization constant

$$1 = \int_0^{\infty} r^2 dr |R(r)|^2 = |N|^2 \int_0^{\infty} r^2 dr e^{-2br} = |N|^2 \frac{1}{4b^3}$$

(copying from the textbook page 435)

This gives

$$N = 2b^{3/2}$$

and the trial wave function

$$R(r)Y_{0,0} = \frac{1}{\sqrt{4\pi}} R(r) = \frac{b^{3/2}}{\sqrt{\pi}} e^{-br}$$

We can get the expectation value as

$$\int d^3r \psi^* \left[\frac{-\hbar^2}{2\mu} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) + \frac{1}{2} \mu \omega^2 r^2 \right] \psi$$

This evaluates to

$$\begin{aligned} E &= 4b^3 \int_0^\infty dr \left(-\frac{\hbar^2 b^2}{2\mu} r^2 + \frac{\hbar^2 b}{\mu} r + \frac{1}{2} \mu \omega^2 r^4 \right) e^{-2br} \\ &= \frac{\hbar^2 b^2}{2\mu} + 4b^2 \int_0^\infty \frac{1}{2} \mu \omega^2 r^4 e^{-2br} dr \\ &= \frac{\hbar^2 b^2}{2\mu} + \frac{3\mu \omega^2}{2b^2} \end{aligned}$$

We minimize the expectation value of the energy,

$$\begin{aligned} \frac{\hbar^2 b}{\mu} - \frac{6\mu \omega^2}{2b^3} &= 0 \\ b^4 &= \frac{3\mu^2 \omega^2}{\hbar^2} \end{aligned}$$

We can plug this back in to find the energy

$$\begin{aligned} E &= \frac{\hbar^2}{2\mu} \frac{\sqrt{3}\mu\omega}{\hbar} + \frac{3\mu\omega^2}{2} \frac{\hbar}{\sqrt{3}\mu\omega} \\ &= \frac{\sqrt{3}}{2} \hbar\omega + \frac{3}{2\sqrt{3}} \hbar\omega \\ &= 1.73 \hbar\omega \end{aligned}$$

The exact form of the ground state energy is

$$\begin{aligned} E &= \left(2n_r + l + \frac{3}{2} \right) \hbar\omega \\ &= \frac{3}{2} \hbar\omega \\ &= 1.5 \hbar\omega \end{aligned}$$

These two are off by 0.23, but our approximation is close.